

Transforming Emotional Health: Smartwatches as Catalysts for Emotional Regulation and Self-Awareness

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Abstract

The rapid progression of wearable technology, particularly smartwatches, has reshaped daily activity engagement and health monitoring. The psychological effects, especially on emotional regulation and self-awareness, are less studied. This research examines how interactions with smartwatches affect emotional regulation and self-awareness, using a 14-day digital diary from 75 participants. By tracking daily smartwatch use and emotional reactions, the study reveals psychological advantages such as stress reduction, enhanced positive emotions, and increased reflectiveness. Features such as health monitoring, activity alerts, and mindfulness activities significantly contribute to these benefits. This study highlights smartwatches' role in not just health tracking but also in building emotional resilience and self-awareness, bridging physical and psychological well-being, and integrating wearable technology into holistic practices.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Empirical studies in interaction design**; **Empirical studies in ubiquitous and mobile computing**; **Interaction devices**; • **Applied computing** → **Psychology**; **Health informatics**.

Keywords

Emotional Regulation, Self-Awareness, Smartwatch Interaction, Wearable Device, Diary Study

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1 Introduction

The rapid advancement of wearable technology, particularly smartwatches, has revolutionized how individuals engage with their daily activities and monitor their physical well-being. While these devices are primarily marketed as tools for tracking health-related

metrics such as heart rate, steps, and sleep patterns, their potential to influence psychological well-being remains underexplored. Smartwatches, with their ability to provide real-time feedback and personalized interventions, offer a promising avenue for enhancing emotional regulation and self-awareness. Emotional regulation, the process by which individuals modulate their emotional responses to challenges, and self-awareness, the conscious recognition of one's own emotions and behaviors, are critical components of mental health and well-being. Existing research indicates that smartwatches can play a significant role in emotional regulation [8, 31, 32] and self-awareness [21, 22, 32]. However, much of the current literature focuses on narrow use cases. There remains a need for more comprehensive studies that explore how everyday smartwatch interactions influence emotional regulation and self-awareness across a broader range of users.

The integration of technology into these psychological processes presents a unique research opportunity to understand how wearable devices can assist in managing emotions and promoting self-awareness [28, 29]. Despite the widespread use of smartwatches, there is a notable gap in research examining the psychological implications of these devices, particularly in the context of emotional regulation and self-awareness. Previous studies have explored the physical health benefits of smartwatches [8, 19, 26], but few have investigated their impact on mental health. This study aims to address this gap by exploring how interactions with smartwatches influence users' emotional regulation and self-awareness, and how these devices are perceived as tools for enhancing these psychological aspects. As such, the research questions for this study are as follows.

- **RQ1:** How do interactions with a smartwatch influence users' emotional regulation?
- **RQ2:** How do interactions with a smartwatch influence users' self-awareness?
- **RQ3:** How do users perceive the effectiveness of a smartwatch in enhancing emotional regulation and self-awareness?

This study aims to fill the existing gap in the literature by examining users' perceptions of smartwatches as tools for enhancing psychological well-being and informing the design and development of more psychologically supportive wearable technologies, focusing on emotional regulation and self-awareness to highlight the broader psychological implications, thereby contributing to technology-driven mental health interventions.

2 Methodology

The study employed a 14-day digital diary method to investigate the psychological benefits and impact of daily smartwatch interactions on emotional regulation and self-awareness. Participants were

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allowed to document their experiences, with minimal researcher interference. Data was collected on an online platform with flexible response types. This method effectively captures immediate reflections, supports authentic self-reporting, and reduces recall bias by allowing participants to document their experiences immediately and honestly, making it highly suitable for studies focused on personal insights, emotional regulation, and continuous personal engagement with smartwatches. Furthermore, it provides a comprehensive dataset for reflecting on how smartwatch interactions affect emotional awareness and self-regulation in diverse daily contexts [5, 10, 12, 15, 30].

2.1 Diary Design and Procedure

Participants were instructed to document their smartwatch usage and reflect on how these interactions influenced their emotional states and personal awareness. This approach allowed for the capture of in-depth and nuanced insights, supporting the analysis of evolving patterns over time. Participants were recruited through Prolific¹, a crowd-sourcing platform chosen for its ability to provide diverse, high-quality samples and its strong participant reliability, as evidenced by prior research [24, 25]. Google Forms was integrated with Prolific to facilitate daily diary entries, and participants received daily reminders and monetary compensation through Prolific to encourage consistent participation. The average completion time for entries was approximately 5 minutes, with participants compensated £0.75 per day, due to participants should complete two mandatory pre-screening questions with a single answer and three compulsory main questions with single, multiple, and free-text answers.

To begin, the participants provided their informed consent by entering their Prolific ID and responding to initial screening questions: “*Did you wear a smartwatch today?*” and “*Did you use any of the following smartwatch feature(s) today?*” Only participants who confirmed wearing a smartwatch and participating in at least one feature were eligible to complete the daily diary entry. Participants reported the specific smartwatch feature(s) they used, their corresponding emotional responses toward the feature used, and their interaction with the smartwatch feature and emotional reactions. Table 1 outlines the diary questions, response types, and response choices used for each daily entry.

2.2 Participants

We pre-screened and recruited participants via the Prolific platform, focusing on specific criteria: owning a smartwatch, frequent use, an interest in joining a two-week diary study, and the capacity to submit daily entries. Initially, we enlisted 100 participants, but only 75 completed the 14-day study. Participants received a comprehensive overview of the study objectives and gave their informed consent. Data were collected anonymously, and withdrawal from the study was allowed at any time without repercussions. Prolific sent daily reminders to motivate participants to complete their diary entries.

The study involved a total of 75 individuals, with 40 (53.33%) identifying as male and 35 (46.67%) as female. A significant portion of the participants hailed from the United Kingdom (31; 41.33%), followed by those from South Africa (8; 10.67%), Poland (7; 9.33%), the United

States (6; 8.00%), Mexico (4; 5.33%), and both Italy and Portugal with 3 each (4.00%). Greece, Hungary, and Norway each contributed 2 participants (2.67%). In addition, one participant (1.33%) came from each of the following countries: Chile, Canada, Bangladesh, the Russian Federation, Zimbabwe, the Netherlands, and Spain. Regarding age distribution, 63 individuals (84%) were young adults aged 19 to 39, 11 (14.67%) were middle-aged adults aged 40 to 64, and there was one older adult (1.33%) aged over 64.

Among the smartwatch brands, Apple was the most popular, with usage reported by 22 participants (29.33%). This was followed by Samsung with 14 users (18.67%), Fitbit with 11 users (14.67%), Garmin with 7 users (9.33%), and Huawei with 5 users (6.67%). Additional brands mentioned were Xiaomi, Amazfit, Rubicon, Aptkdoe, Bounce, Sterns Volkano, Reflex Active, Astrum Peripherals, Google, and Hero.

2.3 Data Analysis

We analyzed direct quotes from participants to interpret our findings and dialogue through thematic coding of open-ended responses. We applied an inductive thematic approach, which is optimal for exploratory research and identifying emergent themes or patterns. This technique highlights the complexities and subtleties of contexts specific to smartwatch functionalities [4, 18, 33]. Feedback from participants was synthesized into themes or patterns that informed our findings and discussions.

Before starting the analysis, a comprehensive evaluation of the data quality was conducted. A total of 1050 diary entries were gathered from 75 participants throughout the 14-day study. The coding process involved several phases. Initially, timestamps (Day 1 through Day 14) and participant IDs (P01-P75) were arranged in a spreadsheet to better understand the data for each participant as well as collectively over the study period. Subsequently, we utilized the free-text responses (question number 3) from the diary form shown in Table 1 for data synthesis and coding analysis. All responses from the free-text section were imported into ATLAS.ti² in order to systematically review every entry.

Throughout the coding process, we classified all unstructured text inputs into specific codes and categories. We first generated preliminary codes using the AI coding feature in ATLAS.ti. We then refined these codes by clustering similar ones, either by merging them or grouping related codes under a single category. This was based on analyzing participants’ quotes and assigning new labels as necessary. This process resulted in identifying three research questions: emotional regulation, self-awareness, and user perception. In the last step, we performed an iterative review to validate and ensure that all AI-generated codes and quotes were accurately assigned to their respective categories.

3 Findings

We report on participant insights regarding how interactions with a smartwatch influence their emotional regulation, their self-awareness, and their perceptions of the smartwatch’s effectiveness in enhancing both emotional regulation and self-awareness.

¹<https://www.prolific.com/>

²<https://atlasti.com/>

Table 1: Diary entry form

No	Question	Answer Type	Answer Choice
1	What feature did you use today?	Single answer	Alarm, Body Composition, Blood Oxygen/ Blood Pressure, Calendar, Call/ Phone, ECG, Heart Rate, Message, Music, Notification, Reminder, Statistics of kcal/ Burn, Sleep, Stress, Weather, World Clock, Workout/ Exercise, Other (write the feature)
2	Feeling or reaction using that feature.	Multiple answer	Excited, Delighted, Happy, Pleased, Content, Relaxed, Peaceful, Calm, Tired, Bored, Depressed, Disappointed, Frustrated, Angry, Tense, Annoyed
3	Please share in more detail all interactions you did with your smartwatch today and your feelings/ reactions.	Free-text answer	

3.1 RQ1: The Influence of Smartwatch Interactions on Emotional Regulation

Participants experienced the dual advantages of smartwatches, which minimized mental interruptions while enhancing positive emotional states. The interplay of these competing and complementary effects highlights the importance of smartwatches in mitigating cognitive load and promoting emotional well-being.

3.1.1 Mental Interruption Reduction. Participants emphasize the role of smartwatches in regulating emotional states and minimizing mental distractions. Smartwatches facilitate task focus by providing features such as activity tracking, seamless notification delivery, and automated tasks that could disrupt cognitive or emotional processes. Motivational reminders and progress tracking help participants manage frustration and stress, providing emotional stability in challenging situations, while reliable features like smooth audio quality during calls enhance focus and emotional clarity by reducing disturbances.

"The smart watch started recording my walk automatically and let me know when I met my daily and weekly targets I like that my Apple Watch allows me to track how my exercise is going. It warns me when I hire some mile stones (such as actual miles, time, steps)."

"Throughout the day, it effortlessly delivered notifications, allowing me to stay connected without constant distractions. Whether it was managing my schedule, controlling my music, or monitoring my fitness, my smartwatch has become a real blessing."

3.1.2 Positive Emotions. Participants indicate that smartwatch aids in emotional regulation by enhancing positive emotions through diverse functionalities. These encompass the promotion of a sense of achievement, the enhancement of self-awareness, and the establishment of a cohesive connection with others. Smartwatches significantly contribute to emotional well-being through progress tracking, providing motivational feedback, and facilitating daily convenience.

"I like using the automatic sleep tracker on my smart watch, and it was extra important for me today because I usually have trouble falling asleep, but last night I

didn't. I also got more sleep than usual last night (I usually only get about 4-5 hours between being unable to fall asleep, and waking up early and not being able to fall back to sleep), and last night I didn't have those issues, so it was nice to see the difference in sleep quality, time, and the effect that it has on my other health aspects in comparison to my regular sleep routine."

"I like my watch notifies me of any messages I get. It automatically starts a walk for me when I go walking. It keeps track of my heart rate and the ECG feature. I also like to see how many steps I've done in a day and how many calories I've burned."

3.2 RQ2: The Impact of Smartwatch Interactions on Self-Awareness Enhancement

We highlight how participants perceive the roles of mindfulness and self-control as mediated by smartwatches, highlighting the interaction between these dimensions to foster self-awareness. The participants' reflections reveal how these devices contribute to enhanced discipline and personal growth by promoting a foundation of awareness.

3.2.1 Mindfulness and Awareness. Participants indicate that smartwatches significantly contribute to mindfulness by offering reminders, monitoring habits, and increasing awareness of daily routines. This indicates a growing dependence on smartwatches to address deficiencies in self-regulation and awareness. In addition, participants often become aware of their habits or behaviors, typically prompted by a lapse in attention or a particular event. This reflective component improves understanding of individual patterns, highlighting the importance of self-awareness in personal development.

"I like that my watch has automated different alarm times for different days. It rings at a different time on weekends as compared to weekdays and also gives me various reminders during the day."

"Unfortunately, I had to download it later because it needed to be recharged. I like it keep an accurate record when im exercising and check the result and make sure its working right and doing its job. It was good to be

able to check the calendar and see what events I had on. Also someone asked me the date and I could quickly open the calendar app from my home screen and let them know the exact date."

3.2.2 Self-Control. Participants report that smartwatches facilitate self-control by providing timely reminders, reminders, and feedback. This external support facilitates adherence to routines, minimizes distractions, and maintains discipline. Participants show deliberate attempts to alter their behavior, frequently motivated by data or feedback generated by smartwatches. These efforts encompass the management of dietary habits, the adherence to exercise routines, and the regulation of sleep patterns. The feedback loops offered by smartwatches enhance accountability, allowing participants to establish and maintain effective routines. This strengthens habits and minimizes instances of self-control failure.

"I have been making sure I wear my watch as much as possible so that my steps are tracked. I am trying to lose weight and my trainer has given me a goal of walking 10k steps, wearing my watch keeps me on track and helps me reach my goal. I like that it notifies me when my heart rate hikes and always wakes me up or never loses time I like that my watch recording my steps in work."

"I used my smartwatch today to do the exercise. I wanted to count the calories I burned. I wish there was a better option for gym training, not just functional training, but I enjoy using it anyway. I felt satisfied and I can have my diet under control."

3.3 RQ3: Perceived Effectiveness of Smartwatches in Supporting Emotional Regulation and Self-Awareness

Participants expressed that smartwatches can improve personal efficiency and emotional well-being, but also support alignment with personal goals and adaptability to individual lifestyles. They reported contrasting experiences with these devices, highlighting their dual role as functional tools and emotionally resonant aids in modern life.

3.3.1 Connectedness and Personalization. Participants indicate that smartwatches enhance connection to personal goals, facilitate lifestyle adjustments, and improve interactions, while also seamlessly integrating into daily routines through personalized features that adapt to individual needs. Individuals value the ability to customize features, including notifications and alerts, to align with their individual lifestyles and objectives. Participants characterize their smartwatch as more than a mere tool, comparing it to a "partner" or "assistant" that aids in managing various tasks.

"I struggle with time management, even on non work days when I have tasks to do around the house. My watch along with reminders and notification help me stay on task and give me nudges to do particular jobs that then get done. Having it always on me means I get the notifications no matter where I am in the house, as I normally leave my phone in other rooms."

"Its easy and very usefull to be able to answer a call without having to have a phone on me. I really like the alarm functions in my smartwatch. It wakes me up with no problem and I never fall asleep to work. I love the notification features. I don't miss anything important and I don't have to use my phone. It makes my life easier and saves me time."

3.3.2 Positive Perception. Participants demonstrate positive attitudes towards their smartwatches, consistently indicating satisfaction with the variety of functionalities offered, including fitness monitoring, notifications, and health monitoring. Positive feelings arise from multiple factors, such as beneficial features, improved convenience, and improved quality of life. Participants recognize that their smartwatch facilitates the management of daily tasks, enhances communication efficiency, and ensures access to information. The smartwatch is perceived as a dependable tool that participants can trust to meet their needs efficiently.

"I love the different features that my watch had. I like that I can start counting steps as soon as I wear it, which also allows me to meet my daily walking limit. I like how I am able to answer calls especially when I'm on the treadmill. Lastly, my smart watch helps keep me motivated to remain active regularly. I like that my watch wakes me up intelligently in the morning because it tracks when I'm in a deep sleep period. After that I can easily check my sleep on the watch, next I use the exercise function to do my morning workout."

"I like that my smart watch tracks all of my exercise and tracked all my steps from a long walk today. It also tells my how many calories I've burned today and my heart rate. It's great for me to be able to track all this at the minute as I'm trying to get fitter and lose a bit of weight. It really helps to keep me motivated when looking at all my stats and trying to improve them each week."

4 Discussion

The findings underscore the importance of designing wearable devices that go beyond physical health monitoring to include features specifically aimed at psychological well-being. We discuss our key findings from user experiences with smartwatches as tools for improving emotional regulation and self-awareness, and also note the limitations and future work of this study.

4.1 Smartwatches as Tools for Enhancing Emotional Regulation and Self-Awareness

Smartwatch engagement significantly influences users' emotional regulation and self-awareness, with users generally perceiving these devices as effective tools for enhancing both aspects. Key insights derived from user experiences are presented below.

Reflective Progression. Users valued tracking health indicators because it helped them manage stress and identify fatigue levels for emotional well-being. The ability to set reminders and track personal goals increases self-awareness. Users often set daily goal steps and feel accomplished when they reach them. This objective-driven

technique encourages movement and health reflection. Digital technologies can improve emotion control and reduce stress through practical feedback and emotional awareness, which makes them important for monitoring affective reactions. Goal-oriented wearable devices can monitor physiological markers and provide feedback to encourage health behavior adjustments and reflective assessments of activity levels and general health [2, 31, 32]. These findings support user perceptions of increased control, relaxation, motivation, and accomplishment in stressful conditions.

Notifications and Connectivity. The ability to receive notifications and manage tasks directly from a smartwatch enables users to maintain connectivity without the need to frequently check their phones. This may alleviate anxiety about the potential of missing important messages or calls, as users reported feeling relieved by the ability to read messages while participating in other activities. Rapid response to messages enhances a sense of control and mitigates feelings of overwhelm. This feature enables users to remain aware of their commitments and responsibilities during the day. A study emphasizes the role of wearable technology in improving emotional well-being by reducing cognitive load and providing timely information [29]. This supports user feedback on the reduction of overwhelm and the management of commitments and responsibilities during the day.

Encouragement for Physical Activity. Numerous users report that their smartwatches serve as a motivation to maintain physical activity, correlating with enhanced mood and emotional regulation. Reminders to engage in physical activity or notifications regarding the achievement of step goals can enhance user motivation and satisfaction. Physical activity is associated with better emotional health and is documented to reduce stress and anxiety, thus improving self-awareness. Research reveals a significant correlation between physical activity, stress relief, and improved mood [13, 20, 22, 26]. In addition, wearable activity trackers significantly improve health outcomes by promoting regular physical activity [1, 3, 16, 17]. This confirms user observations concerning enhanced motivation and satisfaction derived from achieving physical activity objectives.

Mindfulness and Stress Management. Certain smartwatches incorporate functionalities that enhance mindfulness, including guided breathing exercises, reminders to take breaks, and short meditation sessions. Users indicate that these features help with stress management and focus maintenance on busy days, leading to increased self-awareness of their emotional states. In addition to that, users reported that their smartwatch prompted them to engage in deep breathing during a stressful moment, aiding in the restoration of focus. The functionalities play a crucial role in emotional regulation and mindfulness [8, 27], and technology-facilitated mindfulness practices significantly enhance stress management and emotional regulation [11]. This corresponds to user feedback indicating increased focus and reduced stress during busy periods.

Sleep Tracking and Emotional Insight. Sleep-tracking smartwatch users often learn how sleep quality affects their mood and well-being. Users often improve their sleep hygiene by understanding their sleep metrics. Awareness can alter sleep hygiene habits, improving emotional regulation. Some individuals were surprised by their sleep quality and considered changing their habits. Sleep tracking often provides useful information. Physiological data

from smartwatches can improve emotional well-being by revealing sleep patterns, which are crucial for emotional well-being and can lead to behavioral adjustments and better emotional regulation [14, 22, 23, 28, 34]. Sleep tracking can inform users about their sleep patterns and improve decision-making for emotional health.

Interactions with smartwatches enhance users' emotional regulation and self-awareness by providing health monitoring tools, facilitating connectivity, encouraging physical activity and goal setting, promoting mindfulness, and offering insights into sleep patterns. These features collectively enhance the sense of control and understanding of physical and emotional well-being. Furthermore, the psychological implications of smartwatch use include improvements in emotional regulation and self-awareness. Self-tracking behaviors, as discussed by [6, 7], promote reflective practices by prompting users to engage actively with their mental and physical health, while [9] highlights the role of digital technology, particularly wearables, in facilitating mental health interventions and enhancing psychological well-being.

4.2 Limitation and Future Work

While this study offers valuable insights, it is not without limitations. First, the participant pool consisted predominantly of young adults, which may limit the generalizability of the findings to other age groups. Future studies should include a more diverse demographic, incorporating older adults and individuals from various socioeconomic backgrounds to enhance external validity. Second, the relatively short study duration of 14 days may not fully capture longer-term behavioral and psychological changes. Longitudinal research is needed to explore sustained effects and evolving patterns in smartwatch use over time.

Third, the study relied solely on self-reported diary data, which may introduce potential biases, such as recall bias and social desirability effects. Future research should incorporate objective measures, such as physiological or behavioral tracking data, to complement self-reports and provide a more holistic understanding. Additionally, the thematic coding process could benefit from a more detailed explanation of inter-coder reliability and validation, particularly for AI-generated codes. Establishing clear validation procedures and using multiple coders can improve the rigor and credibility of qualitative findings.

Future research should also employ controlled experimental designs to better isolate the effects of smartwatch interactions on emotional regulation and self-awareness. Examining the impact of specific smartwatch functionalities—such as biofeedback, haptic reminders, or AI-driven emotion tracking—could yield deeper insights into their psychological benefits.

The integration of wearable technology into psychological processes presents an opportunity to develop tools that facilitate emotion management and promote self-awareness. By addressing these limitations and expanding research methodologies, we can further understand the transformative potential of wearable technology in advancing mental health and human-centered design. Incorporating technology into psychological processes offers a unique research opportunity to examine how wearable devices facilitate emotion management and improve self-awareness [28, 29].

5 Conclusion

This study underscores the significant role of smartwatches in enhancing emotional regulation and self-awareness, highlighting their potential to improve physical and psychological well-being. Through features such as real-time feedback, health monitoring, personalized reminders, and mindfulness facilitation, smartwatches allow users to regulate key aspects of their health, including heart rate, sleep, and activity levels, while fostering self-awareness through goal setting and reflection. These findings emphasize the transformative potential of wearable devices in mental health practices. By advancing our understanding of the intersection between wearable technology and psychological well-being, this study opens avenues for future research and innovation in human-centered computing, paving the way for digital tools that seamlessly integrate mental and physical health support into everyday life.

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